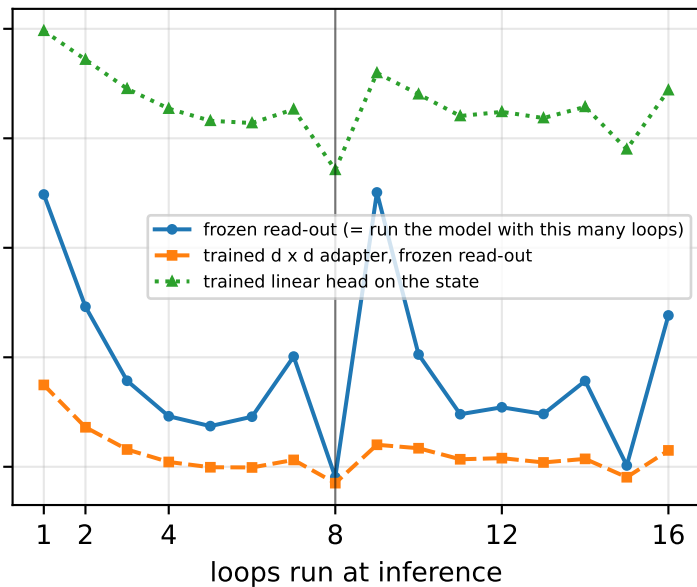


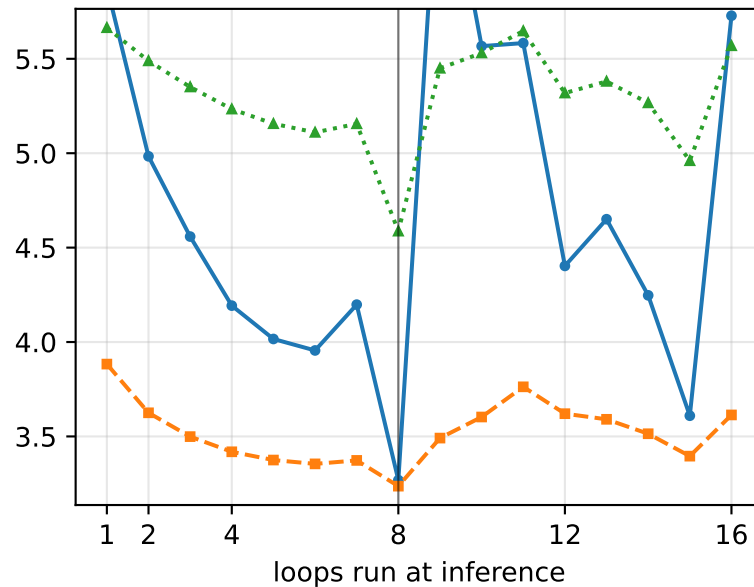
What the state after each loop says about the next token (frozen read-out points above the frame are off-scale; see fig 6)

validation loss (FineWeb-Edu, 2M tokens)

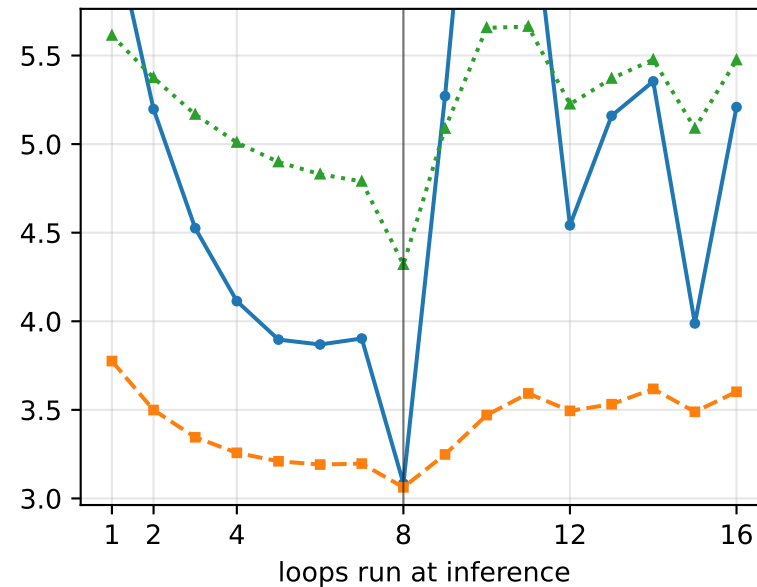
20M middle r8 (trained r=8)



40M middle r8 (trained r=8)



80M middle r8 (trained r=8)



80M whole r8 k4 (trained r=8)

